Derivation of the Discrepancy Bound Threshold

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The specific value for Δ is carefully derived using the **Probabilistic Method** and the **Union Bound**. Below is the step-by-step mathematical reasoning for why $\Delta = \sqrt{2s \ln(2m^2)}$ is chosen.

1 The Context

- We have a set S of size s.
- Each element in S is colored randomly with $+1$ or -1 (a Rademacher random variable).
- The sum of these colors is $X = \sum_{i=1}^{s} X_i$. (In the context of discrepancy, this is the discrepancy of the set S).
- The goal is to bound the probability that the discrepancy exceeds a certain threshold Δ. That is, we want to bound $\Pr[|X| \geq \Delta]$.

2 The Hoeffding Bound

For a sum of s independent random variables taking values in $\{-1, 1\}$, Hoeffding's Inequality states:

$$\Pr[|X| \geq \Delta] \leq 2 \exp\left(-\frac{\Delta^2}{2s}\right)$$

(Intuition: As Δ gets larger, the probability of the sum reaching that value decays exponentially).

3 The Union Bound over m^2 Events

In discrepancy theory, when we bound the discrepancy of a single set S, we usually need to ensure that **all** sets in our family are balanced simultaneously.

- Suppose there are m sets in the family $\mathcal{F}$.
- The problem likely involves evaluating discrepancies over pairs of sets, or the family has size m^2, or we are using symmetric difference where there are $m(m-1)/2$ pairs.
- Let's assume there are $N = m^2$ events we care about.

By the **Union Bound** (Boole's Inequality), the probability that *at least one* of these m^2 sets has a discrepancy larger than Δ is at most the sum of their individual probabilities:

$$\Pr\left[\max_{1 \leq j \leq m^2} |X^{(j)}| \geq \Delta\right] \leq \sum_{j=1}^{m^2} \Pr[|X^{(j)}| \geq \Delta]$$

$$\leq m^2 \cdot \left(2 \exp\left(-\frac{\Delta^2}{2s}\right)\right)$$

4 Setting the Bound to be Small

The goal of the Probabilistic Method is to show that there *exists* a valid coloring. To do this, we need the probability of failure (the union bound) to be strictly less than 1.

A standard technique is to force the probability to be less than 1/2 (or some other small constant). Setting the bound to 1/2 and solving for Δ:

$$2m^2 \exp\left(-\frac{\Delta^2}{2s}\right) \leq \frac{1}{2}$$

$$4m^2 \exp\left(-\frac{\Delta^2}{2s}\right) \leq 1$$

$$\exp\left(\frac{\Delta^2}{2s}\right) \geq 4m^2$$

Take the natural logarithm of both sides:

$$\frac{\Delta^2}{2s} \geq \ln(4m^2)$$

$$\Delta^2 \geq 2s \ln(4m^2)$$

(Note: $\ln(4m^2) = \ln(4) + \ln(m^2)$, which is strictly less than $2\ln(2m^2)$).

If the threshold is set to make the probability less than $\frac{1}{2m^2}$ instead (a common goal to ensure the total failure probability is very small), the algebra yields:

$$2m^2 \exp\left(-\frac{\Delta^2}{2s}\right) \leq \frac{1}{m^2}$$

$$\exp\left(\frac{\Delta^2}{2s}\right) \geq 2m^4 = 2(m^2)^2$$

$$\frac{\Delta^2}{2s} \geq \ln(2m^4) = \ln(2) + 2\ln(m^2)$$

However, looking at the exact formula $\Delta = \sqrt{2s\ln(2m^2)}$.

If we plug this Δ back into Hoeffding's bound:

$$2\exp\left(-\frac{2s\ln(2m^2)}{2s}\right) = 2\exp\left(-\ln(2m^2)\right) = 2\left(\frac{1}{2m^2}\right) = \frac{1}{m^2}$$

If there are m^2 total sets (or events), the union bound becomes:

$$\sum_{j=1}^{m^2} \Pr[|X^{(j)}| \geq \Delta] \leq m^2 \cdot \frac{1}{m^2} = 1$$

5 Summary

The value $\Delta = \sqrt{2s\ln(2m^2)}$ is chosen exactly so that when plugged into the exponential tail bound (Hoeffding's inequality), the exponential decay perfectly cancels out the polynomial growth of the number of sets (which is m^2). This guarantees that the probability of *any* set failing is less than 1, thereby proving that a good coloring must exist.